NSP-233 Name:_____

**Planetary Defense Lab Follow Up**

Please reflect on the following ideas, and write a few sentences under each prompt.

1. Imagine for the moment that an asteroid capable of destroying a large region on Earth is headed right for Paris, France. What ethical and political considerations exist for attempting an asteroid deflection mission? What if France deflects the asteroid just enough that it impacts Munich, Germany? What laws might govern defending an entire planet from an asteroid? What might France do in this situation? What about Germany?

2. The animation we built in the Planetary Defense Simulation lab was two-dimensional – it only considered left, right, up, and down – not in and out of the page. In real life, orbits are three-dimensional, and may be inclined with respect to one another. Additionally, longer period objects like comets may cross Earth's orbit much less frequently that a single asteroid, making characterizing their orbits difficult. Discuss how these considerations might change things. Would you rather be in a position where you need to defend against a comet or an asteroid? Why?